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Magnetic and sequence stratigraphy of redeposited Upper Cretaceous limestones in the Montagna della Maiella, Abruzzi, Italy

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Abstract

The carbonate platform margin of the Montagna della Maiella in the Abruzzi region of central Italy has a complex history of sedimentation, consisting of aggradational and progradational episodes along the shelf margin and pelagic deposition in the basin. A paleomagnetic study at 12 regionally distributed limestone sites of late Cretaceous to Miocene age in the Maiella anticline was unable to define stable magnetic directions at most sites, largely due to the weakness of the magnetizations. At the few usable sites inclinations agreed with reference values based on an African polar wander path. Declinations were inconsistent, indicating local tectonic rotations. Both normal and reversed polarities were preserved at some sites. A magnetostratigraphic section, consisting of 280 samples spaced at 1–1.5 m intervals, was sampled along a profile in the Valle delle Tre Grotte, near the village of Pennapiedimonte. Sequence stratigraphy, based on two-dimensional analysis of outcrops, formed the stratigraphic framework for the magnetostratigraphic study. Rock magnetic analyses showed that magnetite is the dominant ferromagnetic mineral in the limestones. Both alternating field and thermal demagnetization were satisfactory in defining two main components in the weak magnetizations. A component with low coercivity, low unblocking temperature and normal polarity was close in direction to the present axial dipole field direction at the site. A higher coercivity, higher unblocking temperature component with both polarities was interpreted as the characteristic remanent magnetization (ChRM). Although the mean directions of samples with normal and reversed polarity are not exactly antipodal, a stratigraphic sequence of magnetozones is very clearly defined. The Pennapiedimonte magnetostratigraphy correlates well with late Cretaceous Chrons C30r to C34n in the established polarity timescale; numerical ages can thus be allocated to the reversals. Correlation to the Gubbio section allows paleontological dates for the reversal sequence to be inferred. The correlations allow more precise dating of formational and sequence boundaries and a better evaluation of stratigraphic hiatuses. In particular, the boundary between the Tre Grotte Formation (Supersequence 1, Turonian to middle Campanian) and the Orfento Formation (Supersequence 2, upper Campanian to upper Maastrichtian) is shown to be associated with a significant hiatus, spanning most of the middle Campanian *G. ventricosa* zone. Likewise, the absence of Chron C29r at the top of the Orfento Formation (SS2) documents non-deposition during the latest Maastrichtian or

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subsequent erosion. Lower sedimentation rates in the Maastrichtian redeposited sediments than in the Gubbio section suggest that the Maastrichtian record in the Pennapiedimonte section is punctuated by submarine erosion. © 1997 Elsevier Science B.V.

Keywords: paleomagnetism; limestone; Upper Cretaceous; Abruzzi Italy; tectonics; magnetostratigraphy; stratigraphy; correlation

1. Introduction

The Mesozoic and Cenozoic carbonate rocks of the Italian peninsula have proved to be a fertile ground for paleomagnetic investigations. Much of the sedimentary cover is allochthonous and only Apulia, together with the Gargano peninsula, the Iblean zone in Sicily, and small parts of the Southern Alps, are usually regarded as autochthonous. Paleomagnetic data from the autochthons have an affinity with the African polar wander path [1] and are compatible with a tectonic model that ascribes these regions to an Adriatic promontory of the African plate [2,3]. Paleomagnetic directions of limestones in the décollement of the Umbria–Marche area are rotated counterclockwise relative to reference polar wander paths for both the African and European plates. At one time interpreted as paleomagnetic evidence for rotation of the Italian peninsula [4–6], the rotated paleomagnetic declinations evidently represent rotations of the sedimentary cover [7,8]. Differential rotations affecting paleomagnetic vectors probably occurred during the emplacement of thrust sheets [9]. Deviations of tectonic elements and paleo-

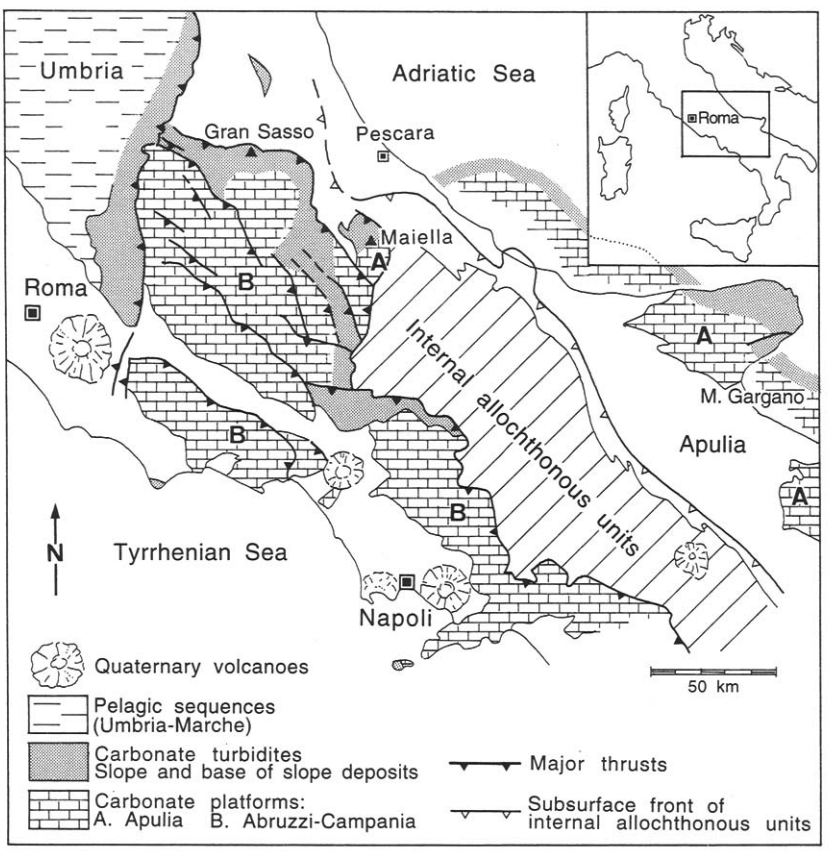


Fig. 1. Simplified map of the Central Apennines showing the main tectonic units, overthrust fronts, upper Cretaceous facies distribution, and location of the Montagna della Maiella.

magnetic directions in the Central Apennines from Adriatic patterns have been tied to Neogene block rotations about vertical axes related to strike-slip faulting [10].

Although a tectonic unit may have undergone rotation relative to reference directions, the rotation can be taken into account in evaluating magnetostratigraphy. Investigations in the pelagic limestones of the Umbrian sequence in the Central Apennines have successfully established polarity sequences that were dated in accompanying biostratigraphic studies. These studies confirmed the record

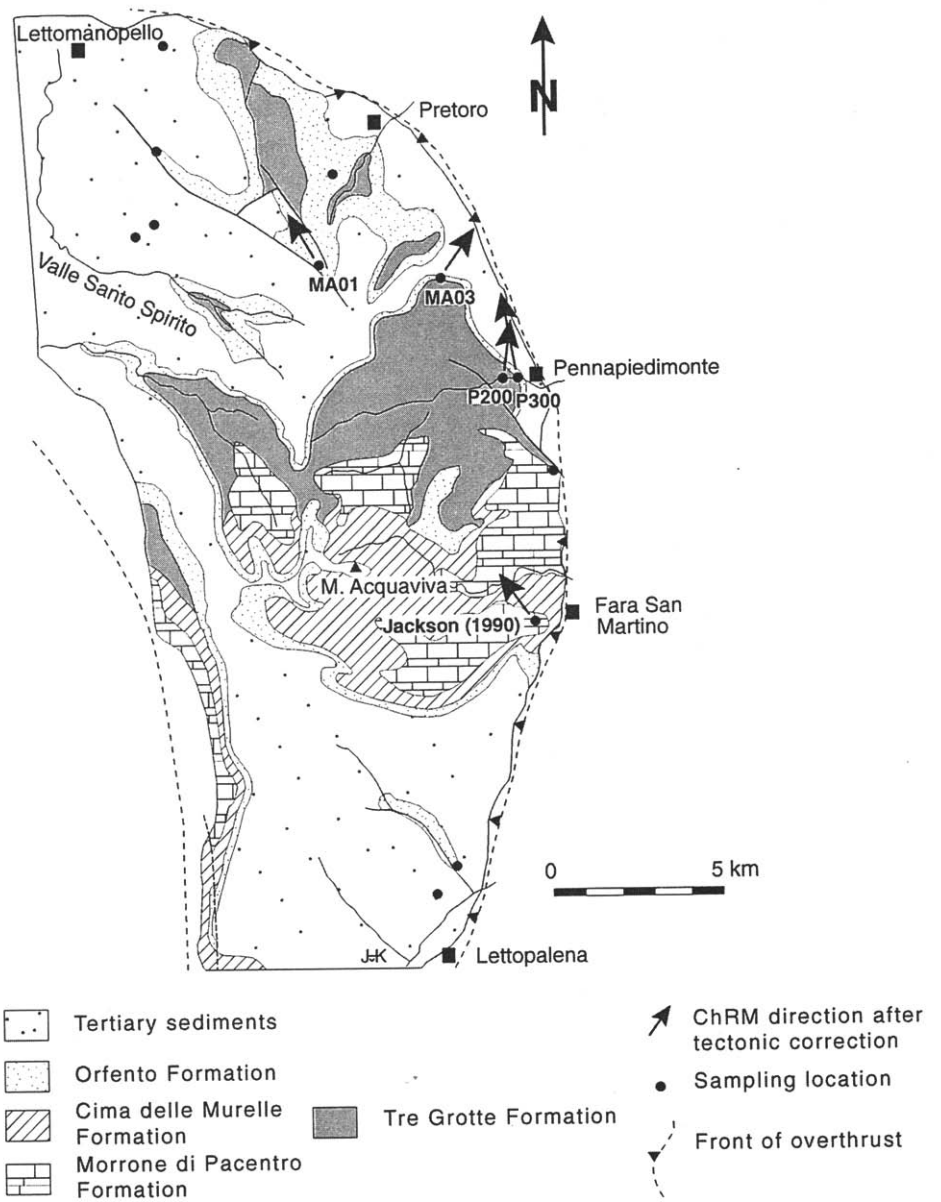


Fig. 2. Schematic geological map of the Montagna della Maiella, showing sampling locations for the regional paleomagnetic study and the site-mean declinations at usable sites, including that of Jackson [25].

of geomagnetic polarity derived from marine magnetic anomalies and tied it to a paleontological framework from the Jurassic/Cretaceous boundary to the Eocene/Oligocene boundary [11]. This makes it possible to use the magnetostratigraphy of a section as a first-order means of dating Paleogene and Cretaceous sediments with high resolution.

In this paper we describe paleomagnetic studies in the Montagna della Maiella tectonic unit in eastern central Italy (Fig. 1). A regional paleomagnetic study was undertaken to complement the tectonic history of the region. A detailed magnetostratigraphic investigation was made in a long continuous section in order to improve the dating of the basinal limestone sequence and to constrain better the duration of the stratigraphic hiatuses and the chronology of depositional sequence boundaries.

2. Geological setting

The Maiella anticline is the frontal part of the most easterly allochthonous unit of the external Apennines in which Mesozoic and Tertiary sediments are exposed (Figs. 1 and 2; [12]). In the late Mesozoic to middle Tertiary the area was situated on the edge of a large, isolated carbonate platform, several hundred kilometres in size (the Apulian platform). The region developed on the southern margin of the Tethys ocean [13]. Sedimentation began in the late Permian to late Triassic with the deposition of variable thicknesses of continental sediments and evaporites [14]. These include the Burano Anhydrite, which later served as a detachment horizon for overthrusting during the Alpine orogeny [14]. Flooding by shallow seas in the latest Triassic to early Jurassic built up the carbonate megabank of the Calcare Massiccio Formation throughout the central and southern Apennines. This megabank disintegrated during early rifting in the 'Liassic', and a number of isolated, Bahamian-type carbonate platforms came into existence [15]. These long-lived carbonate platforms were separated by deeper basins, comparable to the present-day deep troughs of the Bahamian archipelago. They were the site of pelagic and — along the margins — turbiditic and other gravitational carbonate deposition, remote from terrigenous input [13,15]. During most of its early history, at least since the early Cretaceous and up to the late Campanian, the Maiella platform segment was bordered to the north by a steep, non-depositional escarpment (Fig. 3) [16,17], which was most probably inherited from early Jurassic rifting but was shaped during its further evolution by submarine erosion and gravitational collapse.

Towards the end of the early Cretaceous, the Apulian platform became emergent and was karstified. In the Maiella, the exposed bank top was

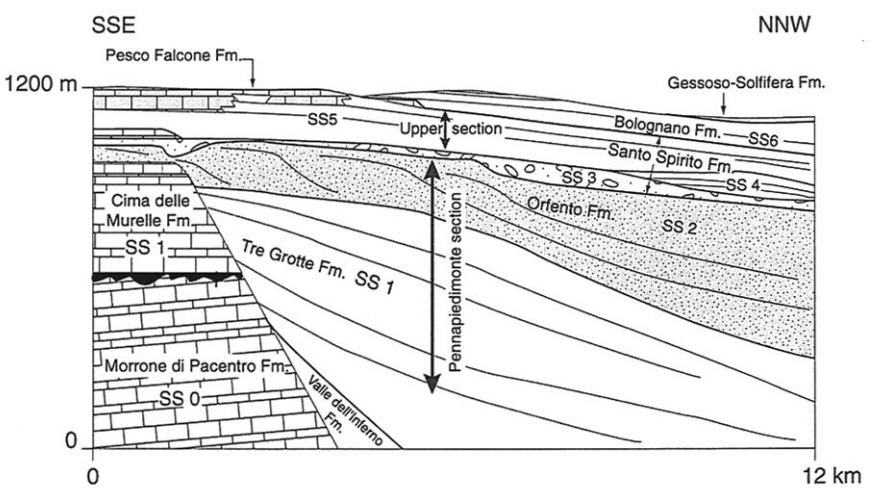


Fig. 3. Schematic representation of the sequence stratigraphy of the Maiella carbonate platform margin and adjacent depositional basin (after Vecsei [18] and Eberli et al. [17]), and approximate coverage of the magnetostratigraphic profiles. Numbers identify Supersequences 1–6.

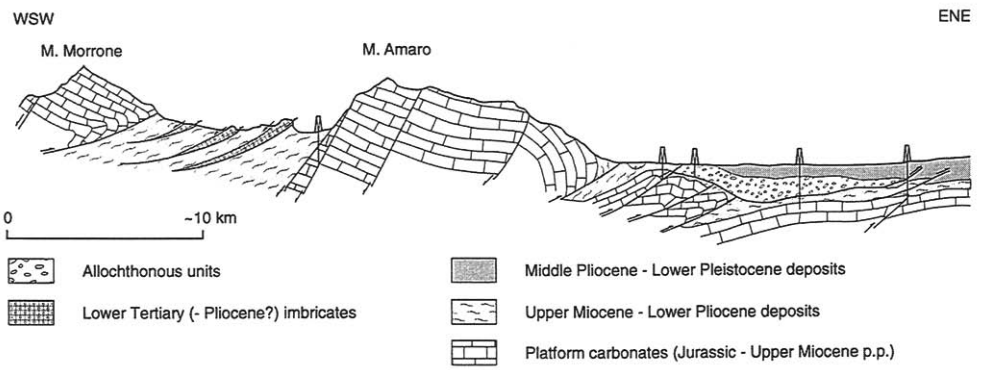


Fig. 4. Tectonic section through the Maiella complex (after Ghisetti and Vezzani [12]).

re-flooded during the Cenomanian [16]. From the Cenomanian to late Campanian a 250 m thick succession of peritidal cycles, bioclastic limestones, and rudist biostromes was deposited on the platform, whereas in the proximal basin a nearly 1000 m thick upper Cretaceous succession of pelagic/periplatform chalks and limestones, carbonate turbidites and megabreccias up to several tens of metres thick onlaps the escarpment (Inferno and Tre Grotte formations [18]; Supersequence 1 of [17]). Because of the higher rate of aggradation in the basin, the deep-water sediments finally buried the escarpment, and shallower facies prograded over the former basin margins in the late Campanian to early Maastrichtian (Orfento Formation, Supersequence 2 of [17–20]). From the latest Cretaceous to Oligocene, phases of subaerial and submarine erosion alternated with phases of prevailing platform aggradation and progradation along this platform margin [17]. A second progradation took place during the late Eocene (Priabonian) to Oligocene (Supersequence 5). It was preceded by a late Paleocene to middle Eocene interval of slope deposition and erosion (Supersequences 3 and 4). In the late Oligocene and early Miocene the Maiella platform was deeply eroded. During the latest Oligocene to early Miocene a temperate carbonate shelf was established in the area (Supersequence 6).

Extensive erosion under subaerial conditions took place during the Messinian, ending the evolution of carbonate platforms in central Italy. During the Miocene–Pliocene the carbonate platforms of the external Apennines were involved in thrusting and folding, resulting in stacked thrust sheets that strike mainly northwest–southeast and verge northeast [14]. The surface expression of overthrusting in the Montagna della Maiella is an anticlinal structure with an axis that has a shallow dip of less than 20° to the north [12]. It is bounded in the west by steep normal faults that set it off from the neighbouring Caramanico depression and in the east by a thrust that separates it from Pliocene deposits and allochthonous units of the Molise basin (Fig. 4).

3. Regional paleomagnetic study

Paleomagnetic samples were taken at 12 regionally distributed sites in the Maiella anticline (Fig. 2). The rugged topography restricted access for paleomagnetic sampling to sites along or near roads in the east and north of the region. At least ten cores, yielding 20–25 samples, were drilled at each site; a total of 255 samples was obtained for the study. The outcrops sampled ranged in age from late Cretaceous to Miocene and represent different lithologies and depositional environments.

The magnetizations were measured with a 2G 3-axis cryogenic magnetometer; the practical noise level of the instrument is equivalent to about 6×10^{-6} A/m. Progressive thermal demagnetization and orthogonal vector diagrams [21] were used to determine the remanent magnetization components in each sample. Unfortunately, the intensity of natural remanent magnetization (NRM) was less than 10^{-5} A/m in about 75% of the collection, so that the demagne-

tized samples were mostly too weak to measure precisely. A reliable ChRM direction could be extracted from only 33 samples at 4 sites in the Tre Grotte and Orfento formations. Two additional sites were formed of groups of 12 samples taken 100 m apart stratigraphically in the Pennapiedimonte magnetostratigraphic section (described below). In some samples a magnetization component with northerly declination and positive inclination was removed below about 300°C (Fig. 5a). This normal polarity component masked a component with negative polarity that persisted up to 500–550°C, which was taken as the ChRM direction. At some sites, after removal of a low blocking temperature component by 200°C, a ChRM with normal polarity was progressively demagnetized (Fig. 5b).

Between 6 and 14 usable samples were obtained at each site. The site-mean ChRM directions and their statistical parameters [22] were computed after inverting reversed polarity samples (Table 1). The mean directions and their circles of confidence are plotted in Fig. 6. Sites MA04 and MA08, with entirely normal polarities, overlap the current axial dipole field direction and may be remagnetized; they were excluded from further consideration. The inclinations measured at the few acceptable sites are fairly consistent at around 40° (±5°), but the declinations show quite large variations (Fig. 2).

Previous paleomagnetic studies on Cretaceous and Paleogene limestones in Central Italy have obtained evidence of tectonic rotations. Declinations at sites in the southwest Apennines showed evidence of both clockwise and counterclockwise rotations of allochthonous décollement nappes during thrusting [23,24]. In an earlier paleomagnetic–tectonic study

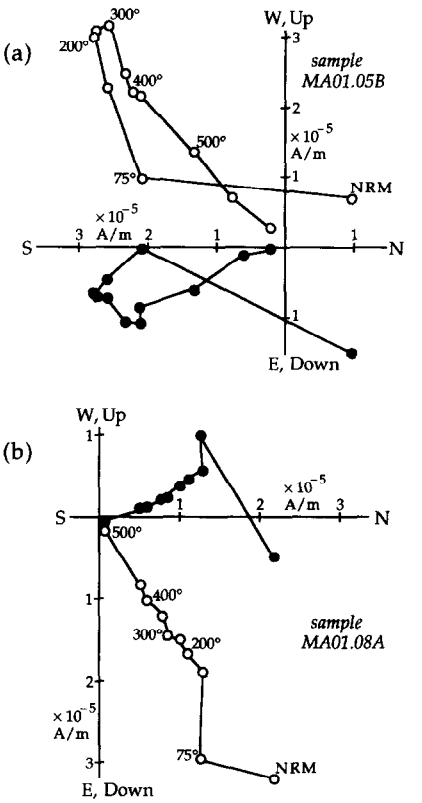


Fig. 5. Progressive thermal demagnetization of samples at a late Cretaceous site, showing ChRM after tilt correction with (a) reversed polarity and (b) normal polarity, after removal of a normal component below 300°C. Numbers beside data points indicate temperature (in °C). Solid symbols = projections on the horizontal plane; open symbols = projections on the N–S vertical plane.

in the Montagna della Maiella, based on limestones of early Cretaceous age, Jackson [25] concluded that the allochthonous Maiella complex had undergone a

Table 1
Summary of ChRM directions (after simple corrections for bedding tilt) at the regionally distributed limestone sites in the Maiella anticline

Site	Age	Formation	N	D	I	α_{95}	k	D_{ref}	I_{ref}	$D - D_{ref}$
MA01	Maastrichtian	Orfento	14	343	44	5	67	345	40	−2
MA03	Eocene	Santo Spirito	7	32	35	10	34	352	44	40
MA04	Early Tertiary	Santo Spirito	6	0	45	6	116	–	–	–
MA08	Latest Oligocene to middle Miocene	Bolognano	6	211	46	10	46	–	–	–
P200	Campanian	Tre Grotte	12	2	40	4	126	341	37	21
P300	Maastrichtian	Orfento	12	345	38	10	19	345	40	0
Jackson (1990)	Early Cretaceous	Morrone di Pacentro	9	326	42	7	44	315	36	11

N = number of samples; D = site mean declination; I = site mean inclination; α_{95} = radius of 95% confidence circle; k = precision parameter [22]. The reference direction (D_{ref}, I_{ref}) for each site is computed from the paleomagnetic data compilation of Van der Voo [26].

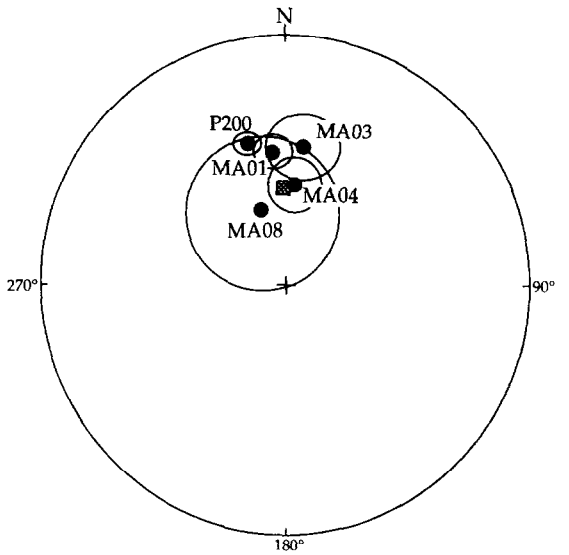


Fig. 6. Lower hemisphere stereographic projection of in situ site-mean directions and their circles of 95% confidence in the Maiella regional study; the square marks the direction of the present axial dipole field.

clockwise rotation of 24° (±15°) about a vertical axis. As in the present case, most of Jackson's samples were unmeasurable. Moreover, the investigation was restricted to a very small area to the west of Fara San Martino, on the edge of the overthrust front (Fig. 2), and so the applicability of the result to the entire Maiella anticline is questionable.

The recent compilation of paleomagnetic data by Van der Voo [26] was used as a basis for computing the reference directions for rocks of early Cretaceous to Miocene age at a location in the Maiella anticline (Table 1). The inclinations observed at the Maiella sites are very similar to the reference inclinations. The declinations at sites MA01 and P300 are not significantly different from the reference declinations; the declinations at sites MA03, P200 and the site of Jackson [25] are rotated clockwise relative to the reference values by different amounts, varying from 10° to 40°. This implies that Jackson's directions are a consequence of local rotation and are not representative of the Maiella crustal block as a whole.

Despite the poor results obtained in the regional study, it is apparent that the upper Cretaceous rocks have good magnetic properties and carry both normal and reversed polarities (Fig. 5). The partially resedimented upper Cretaceous limestones have not as yet been dated in enough detail for high-resolution stratigraphy. A magnetostratigraphic study was carried out with the goal of deriving a polarity sequence, correlating it to the global magnetic polarity timescale, and thereby improving the dating of the depositional sequences recognized in these limestones [17,18].

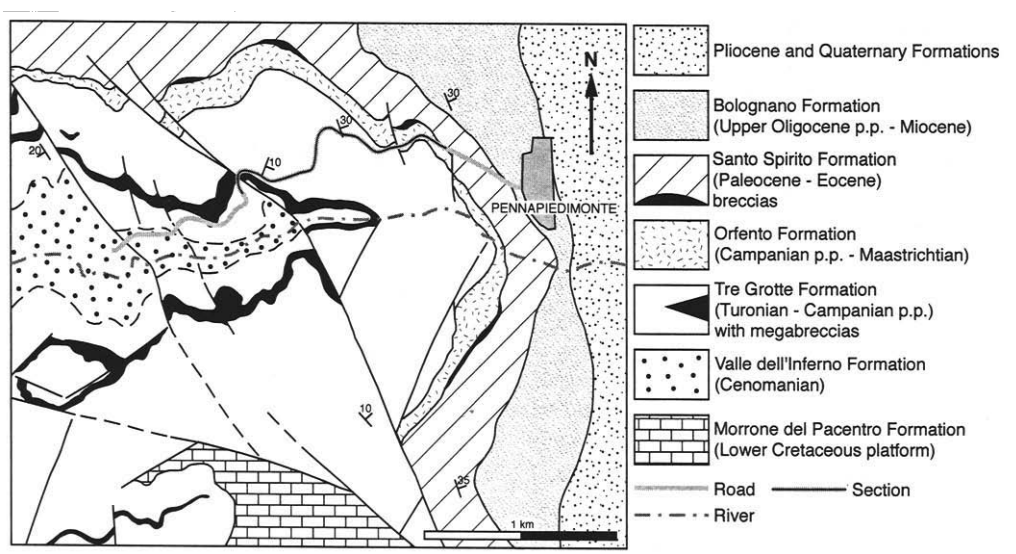


Fig. 7. Geological sketch map of the area west of Pennapiedimonte, Montagna della Maiella (from Accarie [16]), with the location of the section along the road to Valle delle Tre Grotte.

4. Magnetostratigraphy of the Pennapiedimonte section

A great thickness of upper Cretaceous limestones (Supersequences 1 and 2, Fig. 3) is exposed in the canyon-like Valle delle Tre Grotte, which makes an east–west cut through the eastern flank of the Maiella anticline, to the west of the village of Pennapiedimonte (Figs. 2 and 7). A magnetostratigraphic profile was made along a winding dirt road (part of the marked hiking trail No. 4), which provides better access to the head of the valley than an older path deeper in the canyon that it replaces. The stratigraphic framework for the sampling was provided by detailed studies of sequence stratigraphy in the Maiella that preceded our work [17–20].

The sequence stratigraphy was developed from two-dimensional analysis of outcrops [18] and was documented by black and white photographic montages. These were interpreted in the same way as seismic profiles and divided into individual sequences of different order on the basis of combined geometric criteria and facies associations (Fig. 3). Supersequence (2nd-order) boundaries are pronounced truncation surfaces associated with significant chronostratigraphic hiatuses and a change in depositional system, whereas 3rd-order sequence boundaries reflect subaerial exposure of the platform and major events of redeposition in the basin [19]. The sequence stratigraphy was the key to understanding the cycles of progradation and aggradation involved in the sedimentary evolution of the Maiella platform margin [17–20].

In the Valle delle Tre Grotte Supersequences 3 and 4 are reduced to local erosional relics, not present along the profile sampled, and Supersequence 6 was not sampled. Two magnetostratigraphic profiles were measured (Fig. 3). The stratigraphically higher profile was made in Supersequence 5, which, on the basis of paleontological dating with large foraminifera, has a middle Eocene (earliest Bartonian stage) to early Oligocene (earliest Rupelian) age [18]. It was not possible to obtain a polarity sequence in this profile. All 132 samples possessed only a single normal component of magnetization; no reversed polarities were observed. The absence of magnetozones in these rocks, deposited at a time when frequent changes in geomagnetic polarity were taking place, may be due to later remagnetization of the rocks.

The stratigraphically lower magnetostratigraphic profile was made through the upper part of Supersequence 1 (Tre Grotte Formation, Turonian–middle Campanian) and through Supersequence 2 (Orfento Formation, upper Campanian–Maastrichtian). The profile contains truncation surfaces that indicate hiatuses in deposition and submarine erosion. They are usually overlain by breccia or megabreccia beds that are several metres to tens of metres thick. Supersequence 1 can be divided into two parts. The lower part consists of a 240 m thick succession of calcarenites and breccia, with an age estimated as middle to late Cenomanian [16]. These sediments were not sampled. The upper part consists of a thick alternation of upper Cretaceous pelagic limestones, redeposited bioclastic sediments, and breccia containing clasts of platform carbonates. The base of Supersе-

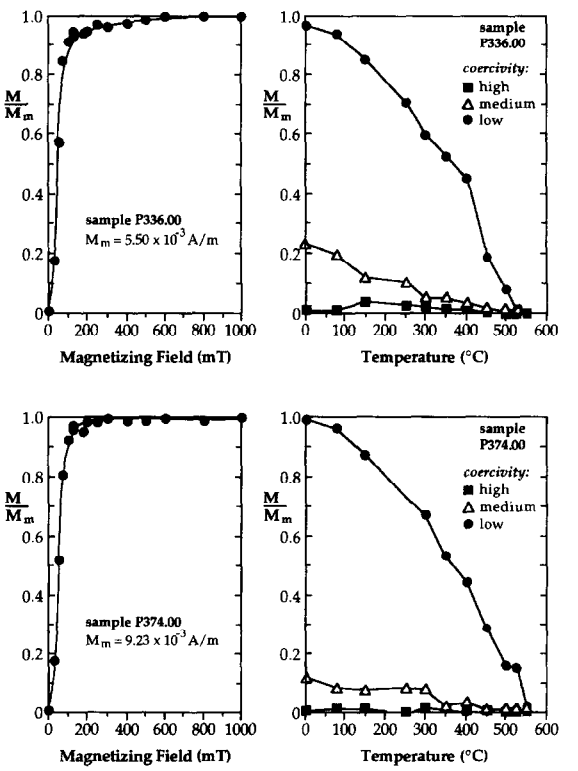


Fig. 8. Analysis of ferromagnetic mineralogy using progressive acquisition of IRM (left) and thermal demagnetization of a composite IRM consisting of mutually orthogonal low (< 0.15 T), medium (0.15–0.6 T) and high (0.6–1 T) coercivity fractions.

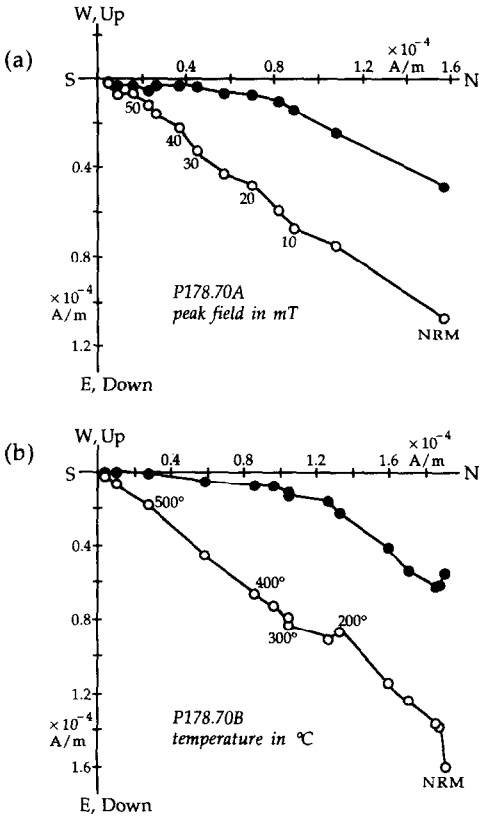


Fig. 9. (a) Progressive alternating field and (b) thermal demagnetization of normal polarity samples in the Pennapiedimonte magnetostratigraphic section. Symbols as in Fig. 5.

quence 2 is a truncation surface with little relief, above which massive calcarenite beds were deposited. The truncation is of late Campanian age and was thought not to represent a significant biostratigraphic hiatus [16,18]. The top of Supersequence 2 is a truncation surface of regional significance. In our measured section it is overlain immediately by Supersequence 5; the hiatus is here of late Maastrichtian to mid-Eocene age [18]. The main Pennapiedimonte magnetostratigraphic profile consisted of 241 samples of pelagic limestones and resedimented calcarenites representing a stratigraphic thickness of 340 m. The rapidly deposited megabreccias, up to tens of metres thick, were not sampled. The sampling interval in the pelagic limestones and calciturbidites averaged about 1.5 m in the lower 100 m, and about 1 m in the upper 240 m of the profile. The indurated limestone offers little resistance to a diamond drill. This 'softness' and the white colour of the limestone raised initial doubts as to whether the ferromagnetic content would be suitable to allow the use of progressive demagnetization techniques.

The ferromagnetic mineralogy of a number of samples spaced at wide intervals along the Pennapiedimonte section was investigated with a method proposed by Lowrie [27]. The samples were first magnetized progressively until a saturation isothermal remanent magnetization (IRM) was achieved. The IRM acquisition curves typically reached saturation in fields below 0.3 T, but in some samples a minor component persisted up to 0.6 T (Fig. 8). A field of 1 T was applied to produce saturation IRM along the axis of the cylindrical test samples. Next, a field of 0.6 T was used to produce a second IRM at

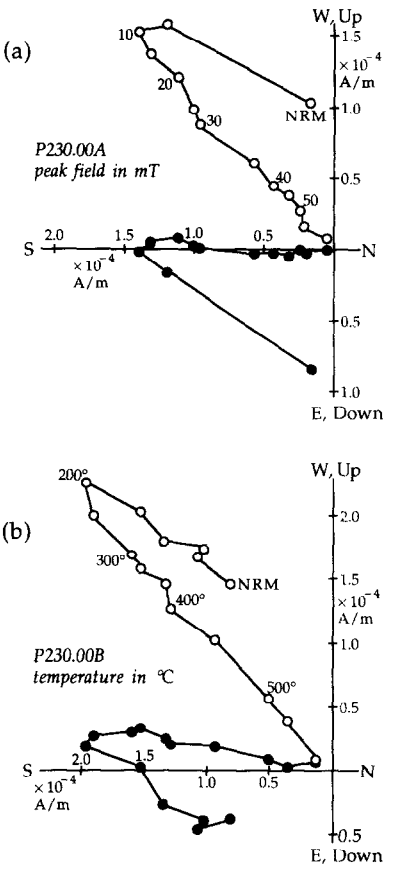


Fig. 10. (a) Progressive alternating field and (b) thermal demagnetization of reversed polarity samples in the Pennapiedimonte magnetostratigraphic section. Symbols as in Fig. 5.

right angles to the first IRM. Finally, an IRM was produced in 0.15 T at right angles to both previous components. The composite IRM thus consisted of mutually orthogonal low (< 0.15 T), medium (0.15–0.6 T) and high (0.6–1 T) coercivity fractions. The thermal decay of each of these fractions was then studied during progressive thermal demagnetization (Fig. 8). The high coercivity component was negligible in all samples. The medium coercivity fraction was largely eliminated by 300–350°C. Both maghemite and pyrrhotite have maximum unblocking temperatures in this range. However, the persistence of a small fraction up to 0.6 T in some IRM acquisition curves suggests that a minor component may be carried by pyrrhotite. In all samples the low coercivity component was dominant; it was eliminated at temperatures in the range 530–580°C and is interpreted as being carried by magnetite.

As in the regional study, the intensity of the NRM was very weak. However, approximately 75% of the samples collected were usable. Pilot studies carried out on twin specimens from the same drilled sample showed that both AF and thermal demagnetization could successfully define two components of remanent magnetization. The orthogonal vector diagrams

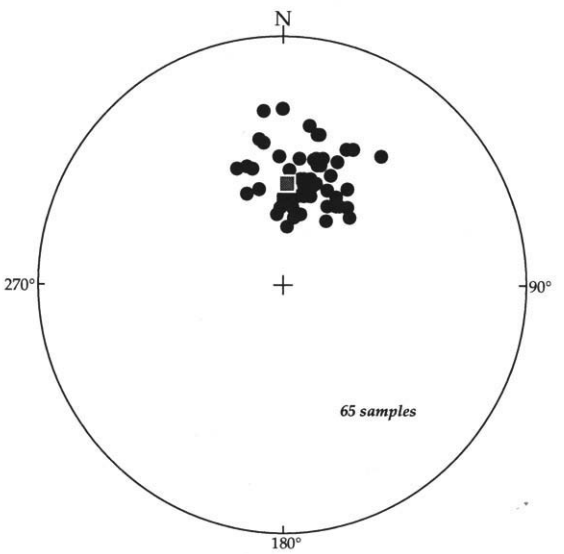


Fig. 11. Lower hemisphere equal-area stereogram showing the directions of the low coercivity, low unblocking temperature component in samples from the Pennapiedimonte section. The square marks the direction of the present axial dipole field at the site.

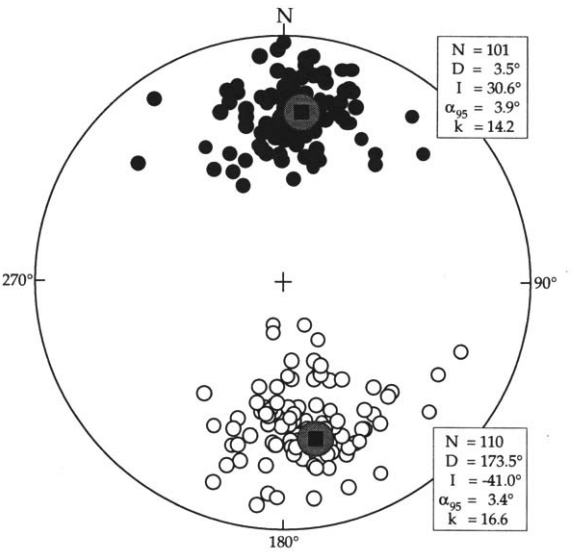


Fig. 12. Equal-area stereogram of the ChRM directions. Dots = plotted on the lower hemisphere; circles = plotted on the upper hemisphere.

[21] show that peak alternating fields up to about 10 mT or temperatures above 200–300°C are sufficient to remove an initial component with normal polarity. It overlies a more resistant component with normal (Fig. 9) or reversed (Fig. 10) polarity, which we interpret as the characteristic remanent magnetization (ChRM) direction of the sample.

The low-coercivity, low blocking temperature, normal polarity component was clearly defined in a large number of samples. The directions of this component group lie around the direction of the present axial dipole field at the site (Fig. 11). We interpret it as a recent overprint, probably of viscous origin.

The ChRM directions form two distinct groups, corresponding to normal and reversed polarities. The means of the respective groups are not exactly opposite (Fig. 12). The angle between the normal mean and the inverted reversed mean is 13.2°; each mean lies outside the circle of confidence of the other. The failure to pass the reversals test suggests incomplete removal of the secondary overprint. Each group contains many directions that deviate by large angles from the group mean. Despite the scatter, the ChRM polarity is unambiguous in each sample.

The declination and inclination of ChRM were

used to compute the latitude of the virtual geomagnetic pole (VGP) during deposition. The latitude of the sampling site during deposition was computed from the inclination of the site mean ChRM; the declination of the site mean ChRM was subtracted from the individual declinations to account for tectonic rotation of the Maiella anticline. The stratigraphic plots of declination, inclination and VGP latitude define a clear sequence of magnetozones (Fig. 13). The profile contains small gaps, corresponding to outcrop-covered intervals and the rapidly deposited megabreccia layers, which were not sampled.

The Pennapiedimonte magnetostratigraphy agrees

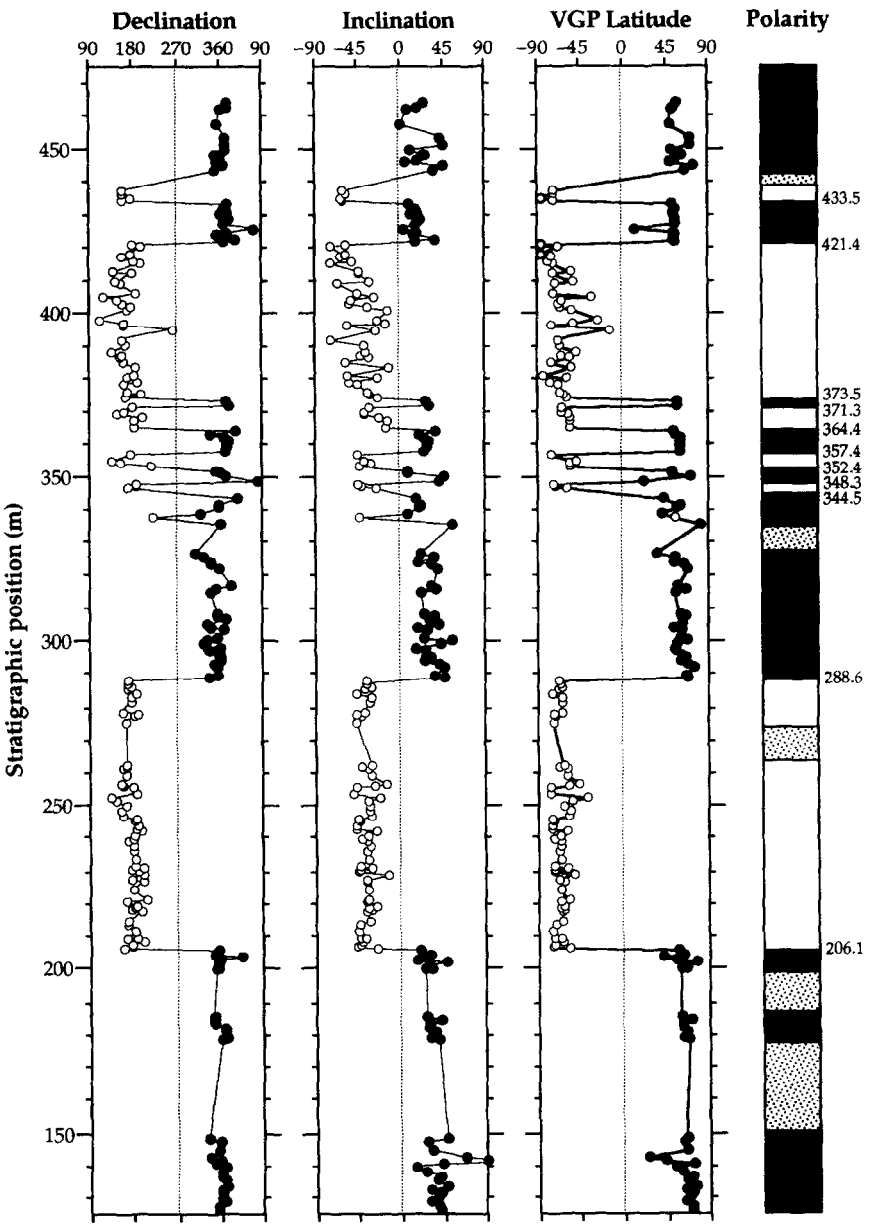


Fig. 13. Stratigraphic plots of declination, inclination and VGP latitude in the Pennapiedimonte section. The margins of magnetozones are defined by the zero-crossing levels of the VGP latitude. Key to polarity column: black = normal; white = reversed; shaded = data gap.

well with other polarity sequences for the late Cretaceous. There is a high degree of correlation with the late Cretaceous part of the revised geomagnetic polarity timescale (Fig. 14a), which was derived from oceanic magnetic anomalies [28]. The section at Pennapiedimonte evidently spans the sequence of polarity chrons from C30n to the top of C34n (the Cretaceous Polarity Superchron, or Quiet Interval). This allows the Pennapiedimonte section to be dated in absolute terms within the accuracy of the reference timescale, by associating a numeric age with each polarity reversal. The correlation gives a mean sedimentation rate of 14 m/Ma in the Pennapiedimonte section. This rate includes deposition of turbidites and other redeposited calcarenites and rudites, but it does not include the thick megabreccias below 200 m (Fig. 15).

5. Formational and sequence boundaries

The Pennapiedimonte magnetostratigraphy also correlates well with magnetozones in the late Cretaceous stratotype section in the Bottaccione gorge near Gubbio [29,30], which is located approximately 190 km to the northwest in the Umbrian sequence (Fig. 14b). The paleontological zonation of the Gubbio section [31] allows the probable levels of biostratigraphic boundaries at Pennapiedimonte to be inferred (Fig. 14b and Fig. 15), so that better dates can be associated with the supersequence (2nd-order) and sequence (3rd-order) boundaries.

Supersequence boundaries are truncation surfaces that also separate different lithological formations, reflecting different depositional systems [17–20]. In our section, the Supersequence boundary (SS1/SS2) separating the Tre Grotte from the Orfento Formation was interpreted as being associated only with a minor hiatus within the *Globotruncana* (now *Globotruncanita*) *calcarata* zone [16,18]. However, our data suggest a much longer duration. The C33r/C33n (A − /B +) boundary coincides with the boundary between the early Campanian *Globotruncana* (now *Globotruncanita*) *elevata* and the middle Campanian *Globotruncana ventricosa* zone (cf. [32]). If present, this zone appears to be restricted to the uppermost few metres of the Tre

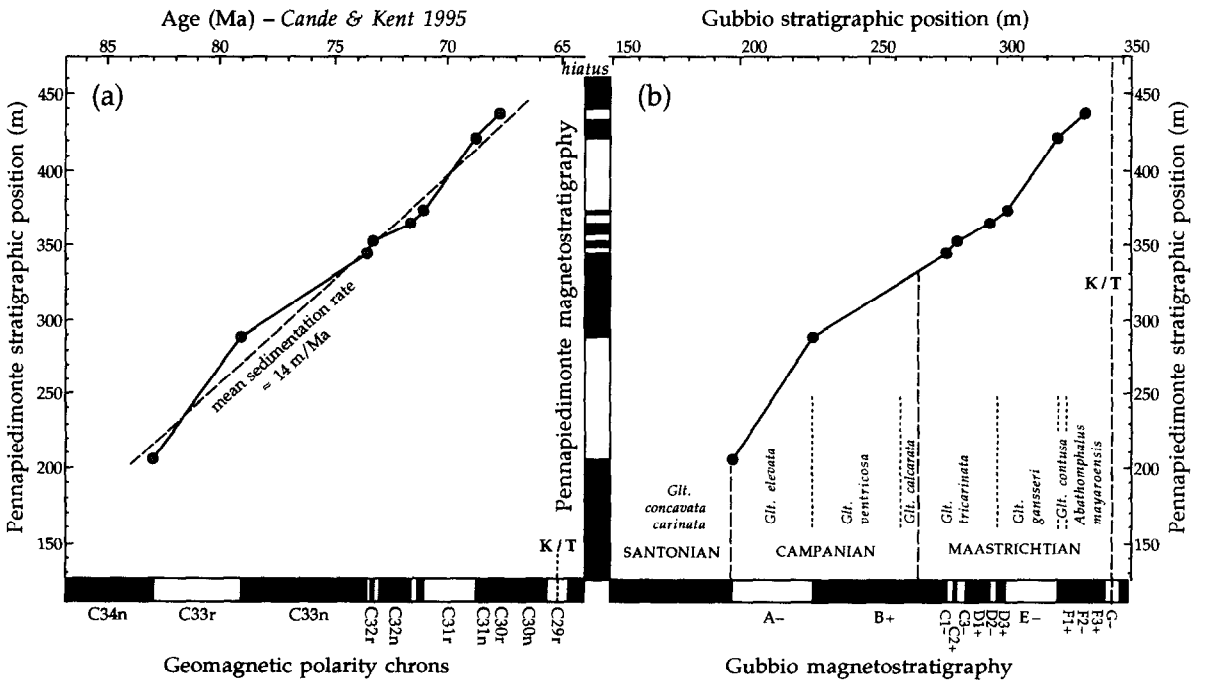


Fig. 14. Comparison of the Pennapiedimonte magnetostratigraphy with (a) the revised geomagnetic polarity timescale for the late Cretaceous derived from oceanic magnetic anomalies [28] and (b) the late Cretaceous magnetostratigraphy of the Gubbio Bottaccione section [29] and its paleontological zonation [31]; *Globotruncana ventricosa* zone added after [32]. Key to polarity columns: black = normal; white = reversed.

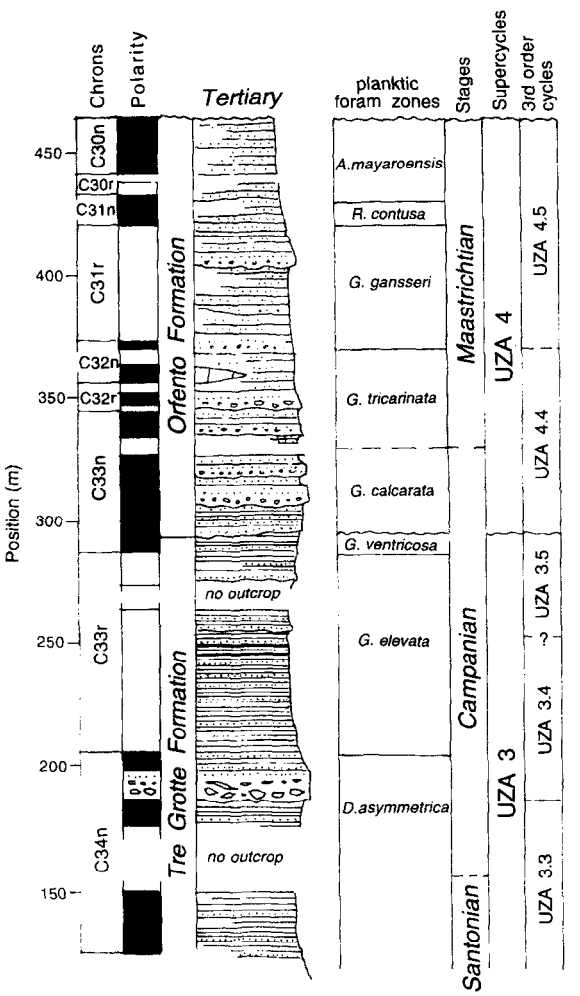


Fig. 15. Tentative correlation of magnetostratigraphy and lithostratigraphy with planktic foraminiferal zonation [31,32] and sea-level cycles [32].

Grotte Formation below the unconformity (Fig. 15). The base of the Orfento Formation is dated by *Globotruncana calcarata* as late Campanian [18]. The resulting hiatus, comprising most of the *G. ventricosa* zone, would also explain why Chron C33n (corresponding to zone B + in the Gubbio section) is thinner in the Pennapiedimonte section than in the generally thinner, distal Gubbio section, which does not contain calcareous turbidites (compare also Chron 33r in the Pennapiedimonte section with zone A − in the Gubbio section, Fig. 14b). A correlation of the supersequence boundary SS1/SS2 with the base of cycle UZA 4.4 of Haq et al. [32] is compatible with our stratigraphic data (Fig. 15; cf. [18,20]). If this is the case, the 3rd-order cycles UZA 4.1 to 4.3 are not represented in the Pennapiedimonte section. This is compatible with the progradation model of Mutti et al. [19], which suggests that, during times of aggradation, sediments were mainly trapped on shelf and marginal areas. However, pelagic sediments deposited during UZA 4.1–4.3 must have been removed by erosion before deposition of sediments corresponding to UZA 4.4 (Orfento Formation).

Supersequence SS2 (Orfento Formation) is overlain along a pronounced truncation surface by relics of Supersequence 3/4 (lower Paleocene to middle Eocene) and Supersequence 5 (middle Eocene to lowermost Oligocene). As Chron C29r (zone G in the Gubbio section) is not present in the Pennapiedimonte section, strata corresponding to the upper part of the late Maastrichtian *Abathomphalus mayaroensis* zone were not deposited or were eroded before deposition of the overlying Paleocene beds. This supersequence boundary might be correlated with the Supercycle boundary UZA 4/TA 1, which also falls in Chron C30n [32].

Lithoclastic megabreccias occurring in the Tre Grotte Formation have been interpreted as lowstand deposits overlying 3rd-order sequence boundaries [17,18]. The megabreccia occurring near the top of Chron C34n falls in the *Globotruncana* (now *Dicarinella*) *concavata carinata* or the *Dicarinella asymmetrica* zone, respectively. Its base could thus be correlated with the base of the 3rd-order cycle UZA 3.4 of Haq et al. [32], The younger UZA 3.4/3.5 3rd-order cycle boundary is, however, not obvious from the stratigraphic record.

Although almost entirely composed of redeposited lithologies, Supersequence 2 shows a lower mean sedimentation rate than the top part of Supersequence 1 (Fig. 14a). Part of the section (Chron C31r) is thicker than the corresponding E − interval of the more basinal Gubbio section, while another part (Chron C31n) is thinner. This suggests that the record in the Pennapiedimonte section is punctuated by erosion (or at least non-deposition). Vecsei [18] distinguished four 3rd-order sequences in this supersequence. One of these sequence boundaries (S 2.3/S 2.4) at the base of the *Globotruncana* (now *Gansserina*) *gansseri* zone could correspond to the UZA 4.4/UZA 4.5) boundary.

6. Discussion

The paleomagnetic directions at the few usable sites in our regional study of the Montagna della Maiella have similar inclinations, which agree closely with expected values derived from reference data for the African plate. This is in agreement with other paleomagnetic observations in the Italian peninsula. The declinations at the usable sites showed quite large regional differences, which probably reflect local tectonic rotations. Considering the errors associated with both the reference data and our paleomagnetic data, our results are compatible with tectonic displacements of the Maiella anticline documented by seismic reflection results [33].

The upper Cretaceous limestones in the Montagna della Maiella record both polarities of the geomagnetic field. Magnetostratigraphy in the Pennapiedimonte section agrees very well with the record of geomagnetic polarity in the late Cretaceous derived from marine magnetic anomalies. The numerical ages of Pennapiedimonte magnetozones can be obtained by correlation to the established polarity timescale. Alternatively, the anticipated levels of major paleontological zones can be obtained by correlation to the Gubbio section. The correlations provide stratigraphers working in the Maiella with greatly improved dating of the resedimented sequence. For example, the ages of minor units within the stratigraphic supersequences can be inferred. The study is an example of how magnetostratigraphy, under favourable circumstances, can provide a framework that permits more detailed basin analysis.

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